

A2: Networks - Exam Lab



This exam practice page is designed for the **A2: Networks** unit of the new IB Computer Science syllabus (first assessment 2027).

Syllabus Links:

A 2.1 Network fundamentals

A 2.2 Network architecture

A 2.3 Data Transmissions

A 2.4 Network security

The Exam Scenario

Scenario: The "Smart-Care" Medical Complex. A large urban hospital has recently upgraded its digital infrastructure. The complex uses a high-speed **Local Area Network (LAN)** to connect surgical robots, patient monitoring systems, and administrative offices. To protect patient privacy while allowing doctors to access records remotely, the hospital employs a **Virtual Private Network (VPN)** and extensive **Network Segmentation**.

Core Questions (HL and SL)

Give these questions a try! If you're stuck, check out the 'Strategy' section to help you get started. Once you're finished, use the Mark Scheme and Worked Example to review your progress. If these feel a bit tough right now, head back to the eBook pages to sharpen your understanding first.

Question 1: Network Fundamentals & Protocols (SL & HL)

- (a) **Identify** one characteristic of a **Personal Area Network (PAN)** that distinguishes it from the hospital's **Local Area Network (LAN)**. [2 marks]
- (b) The hospital's website uses **HTTPS** for its patient portal. **Explain** the importance of the "Secure" aspect of this protocol compared to standard **HTTP**. [3 marks]

Level 1: Strategy

Command Term Check: "Identify" requires a specific answer; "Explain" requires reasons or causes.

The Logic: PANs are about individual devices (meters/phones); LANs are about local buildings. For HTTPS, think about the role of encryption in public environments.

Level 2: Mark Scheme

- (a) Award [1] for identifying short-range (e.g., Bluetooth) and [1] for identifying that it typically serves a single individual rather than an organisation.
- (b) Award [1] for identifying that HTTPS uses encryption (SSL/TLS certificates). Award [1] for stating it prevents unauthorised interception (eavesdropping). Award [1] for mentioning data integrity/authentication of the server.

Level 3: Worked Example

- (a) A PAN is designed for a very short range, often centred around an individual's body (e.g., a doctor's smartwatch connected to their phone), whereas a LAN covers a larger geographical area like a hospital wing and connects many users to shared resources.
- (b) HTTPS is critical because it encrypts the data transmitted between the patient's browser and the hospital server. Unlike HTTP, which sends data in plain text, HTTPS ensures that sensitive medical information cannot be read if intercepted by an attacker. It also uses digital certificates to prove to the patient that they are connected to the genuine hospital portal and not a malicious "phishing" site.

Question 2: Architecture & Segmentation (SL & HL)

- (a) The hospital uses **Virtual Local Area Networks (VLANs)** to separate the guest Wi-Fi from the surgical robot control system. **Explain** one reason why **network segmentation** is used in this context. [3 marks]

(b) **Compare and contrast** the **client-server** and **peer-to-peer (P2P)** networking models for use in a hospital environment. **[4 marks]**

Level 1: Strategy

The Logic: Segmentation is about performance (noise) and security (containment). Client-server is about "Central Control" while P2P is about "Shared Responsibility".

Level 2: Mark Scheme

(a) Award **[1]** for identifying a reason (security/performance); **[1]** for the mechanism (reducing congestion/containment); **[1]** for the context (preventing guests from reaching critical hardware).

(b) Award **[1]** for a similarity (both facilitate data exchange). Award **[1]** for a contrast point on central control. Award **[1]** for a contrast point on scalability/reliability. Award **[1]** for a conclusion on hospital suitability.

Level 3: Worked Example

(a) Network segmentation is used to enhance security by isolating critical systems. By placing surgical robots on a separate VLAN from the guest Wi-Fi, the hospital ensures that even if a guest's device is compromised, the attacker cannot access the medical equipment. Additionally, it improves performance by reducing broadcast traffic and congestion on the surgical network.

(b) Both models allow for data sharing across the hospital. However, a client-server model is superior for hospitals because it provides centralised management of sensitive patient data, ensuring consistent security and backup policies. In contrast, a P2P model lacks a central authority, making it difficult to secure and scale for thousands of users, though it might be more resilient to a single point of failure than a central server.

HL Only : Advanced Networking

Give these questions a try! If you're stuck, check out the 'Strategy' section to help you get started. Once you're finished, use the Mark Scheme and Worked Example to review your progress. If these feel a bit tough right now, head back to the eBook pages to sharpen your understanding first.

Question 3: Data Transmission & TCP/IP (HL Only)

(a) **Describe** how the **Transport layer** and the **Internet layer** of the **TCP/IP model** interact to ensure a medical image is sent correctly across the network. **[4 marks]**

(b) The hospital's IT department must choose between **static routing** and **dynamic routing** for its emergency response tablets. **Outline** one advantage and one disadvantage of using **dynamic routing** in this scenario. **[4 marks]**

Level 1: Strategy

The Logic: TCP/IP layers have specific jobs: Transport (packets/reliability) and Internet (addressing/routing). Dynamic routing is "Smart" but "Complex".

Level 2: Mark Scheme

(a) Award **[1]** for Transport layer segmenting data/adding ports. Award **[1]** for the Internet layer adding IP addresses. Award **[1]** for mentioning the interaction (handing off data). Award **[1]** for mentioning reliability/sequencing (TCP).

(b) Award **[1]** for an advantage (e.g., automatic reconfiguration if a link fails). Award **[1]** for expansion in context. Award **[1]** for a disadvantage (e.g., higher CPU/bandwidth overhead). Award **[1]** for expansion.

Level 3: Worked Example

(a) The Transport layer first segments the medical image into smaller packets and assigns sequence numbers to ensure they can be reassembled correctly. It then hands these segments to the Internet layer, which attaches the source and destination IP addresses. The layers interact as the Internet layer determines the best path for each packet, while the Transport layer (using TCP) monitors for any lost packets and requests retransmission, ensuring the image arrives without errors.

(b) An advantage of dynamic routing is high **resilience**; if a network switch in the hospital fails, the tablets can automatically find a new path to the server without manual reconfiguration. A disadvantage is **increased resource usage**, as routers must constantly exchange information to update their routing tables, which can consume more bandwidth and CPU power compared to a simple static setup.

Question 4: Vulnerabilities & Servers (HL Only)

(a) A hospital server is targeted by a **Distributed Denial of Service (DDoS)** attack. **Explain** how this attack functions and **suggest** one countermeasure the hospital could implement. **[4 marks]**

(b) **Outline** the role of a **Proxy Server** in protecting the hospital's internal network from external threats. **[2 marks]**

Level 1: Strategy

The Logic: DDoS is "Strength in Numbers" (overwhelming the server). A Proxy is a "Middleman" that hides the real server.

Level 2: Mark Scheme

(a) Award [1] for explaining multiple systems (botnets) flooding the server. Award [1] for the result (system crash/unavailability). Award [1] for a countermeasure (e.g., traffic filtering, rate limiting, or DDoS mitigation tools). Award [1] for expansion.

(b) Award [1] for identifying that it acts as an intermediary for requests. Award [1] for mentioning it hides internal IP addresses or filters malicious content.

Level 3: Worked Example

(a) A DDoS attack occurs when an attacker uses a network of compromised computers (a botnet) to flood the hospital's server with a massive volume of requests simultaneously. This overwhelms the server's bandwidth or CPU, making it crash or become unresponsive to legitimate medical staff. To mitigate this, the hospital could use **Intrusion Prevention Systems (IPS)** or specialised DDoS mitigation tools that identify and block abnormal traffic patterns before they reach the main server.

(b) A Proxy Server acts as an intermediary between the hospital's internal users and the Internet. It protects the network by masking the internal IP addresses of the doctors' computers and can be configured to block access to unauthorised or dangerous websites, providing an additional layer of security.

How confident do you feel with this topic?

Exam Ready: I answered most questions correctly without looking at the eBook. I can explain these concepts to someone else and understand the "why" behind the mark scheme.

Move on to the next topic.

Practitioner: I understood the general concepts, but I missed a few marks on specific terminology or complex applications. I needed the "Strategy" guide for the harder questions.

Annotate your answers with the Mark Scheme and revisit the eBook sections where you lost marks. You can also use the bookmarking and notes feature on the eBook pages to highlight certain areas.

Novice: I struggled to get started on the questions. I found it difficult to apply the theory to the practice scenarios.

Go back to the eBook and Worked Examples before attempting these questions again. Perhaps use some of the Quizlets to practice the key terms and review the command terms.

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